SCHOOL OF INFORMATION TECHNOLOGY ADMINISTRATION & SECURITY

CLOUD FUSION SOLUTIONS

Capstone Project Report

CLOUD FUSION SOLUTIONS

Title: Transition of "Fusion Enterprise Security Service" from On-Premises to Serverless Three-Tier Architecture

Guided by: Prof. Pedram Habibi

Team: Group 1

- Pratham Sapra
- Kiranpreet Kaur
- Suma Latha Pittala
- Shrey Anilkumar Patel
- Chaudhary Ashran Haider

Seneca | SCHOOL OF INFORMATION TECHNOLOGY ADMINISTRATION & SECURITY

CLOUD FUSION SOLUTIONS

Table of Content

S

Executive Summary....3
Objective:....3
Key Benefits of Serverless Technologies:....3
Anticipated Outcomes:....3
Introduction to Cloud Fusion Solutions....3
Company Background....3
Mission and Vision....3
Product Portfolio....4
Fusion Enterprise Security Service....4
Overview....4
Key Features....4
Current Infrastructure....5
Migration Goals and Objectives....5
Key Objectives:....5
Alignment with Organizational Strategies:....5
Current Infrastructure Assessment....6
Comprehensive Assessment of Existing Systems....6
Identified Limitations and Proposed Solutions....7
Cloud Service and Tool Selection....7
Justification for Selected Services and Tools....7
Migration Strategy....8
Security and Compliance Strategy....9
Cost Analysis and Budget....10
On-Premises Infrastructure TCO....10
AWS Cloud TCO....11
Savings and ROI....11
Project Management and Timeline....11
Risk Assessment and Mitigation....12

Cloud Architecture Diagram..13
Implementation... 14

Executive Summary

Objective:

The primary goal of this project is to transition Cloud Fusion Solutions' "Fusion Enterprise Security Service" from its current on-premises setup to a serverless three-tier architecture on AWS. This transformation aims to harness the benefits of cloud computing to enhance system performance, improve security, and achieve cost efficiency.

Key Benefits of Serverless Technologies:

1. **Cost Efficiency:** Serverless computing ensures that the company pays only for the compute power used, eliminating the need for over-provisioning and reducing overall costs.
2. **Improved Performance:** Serverless architecture provides automatic scaling, which ensures that the system can handle increased loads seamlessly, resulting in improved performance and user experience.
3. **Enhanced Security:** With built-in security features and automatic updates, serverless services offer a higher level of security compared to traditional on-premises setups.

Anticipated Outcomes:

- **Adaptability:** The new architecture will allow Cloud Fusion Solutions to quickly adapt to changing business needs, fostering innovation and rapid deployment of new features.
- **Reduced Operational Overhead:** By eliminating the need for managing physical servers, the company can focus more on development and innovation.
- **Innovation:** The move to a cloud-based, serverless architecture will enable faster implementation of new features and enhancements, keeping the company competitive in the market.

Introduction to Cloud Fusion Solutions

Company Background

Cloud Fusion Solutions is a leading IT solutions provider, renowned for its extensive product portfolio and expertise in both on-premises and cloud-based services. With a robust workforce of approximately 1200 employees, the company has established a significant presence in the IT industry by assisting organizations in their digital transformation journeys, particularly in migrating to the cloud.

Mission and Vision

Mission: To deliver innovative and reliable IT solutions that empower businesses to achieve their full potential by leveraging the power of cloud computing and advanced technologies.

Vision: To be a global leader in cloud solutions, driving digital transformation and creating value for businesses through exceptional customer experiences, operational efficiency, and continuous innovation.

Product Portfolio

Cloud Fusion Solutions offers a diverse range of products tailored to meet the specific needs of various industries:

1. **E-commerce Solutions: Fusion Cart**
    - A comprehensive e-commerce platform that supports online retail operations, providing features like product management, payment integration, and customer support.
2. **Payroll Solutions: Paystream**
    - An efficient payroll management system designed to streamline payroll processing, tax calculations, and compliance with regulatory requirements.
3. **Banking Solutions: Fusion Pay**
    - A secure and scalable banking solution that facilitates transactions, account management, and financial services for banking institutions.
4. **Security Solutions: Fusion Enterprise Security Service**
    - A critical event management and enterprise safety application that offers a unified platform for incident preparedness, risk monitoring, critical event management, and service reliability.

Fusion Enterprise Security Service

Overview

The Fusion Enterprise Security Service (FESS) by CloudFusion Solutions is a comprehensive, integrated platform designed to ensure the safety, security, and operational continuity of enterprises. It offers a unified approach to incident preparedness, risk monitoring, critical event management, and service reliability, enabling organizations to effectively manage and mitigate risks in real-time.

Key Features

1. **Critical Event Management:**
   - Facilitates coordinated responses to critical events through a centralized command and control interface.
   - Enables real-time communication and collaboration among response teams.
   - Supports incident documentation and post-event analysis for continuous improvement.
2. **Service Reliability:**
   - Ensures high availability and reliability of security services through robust infrastructure and failover mechanisms.
   - Includes automated backups and disaster recovery plans to safeguard against data loss and service disruptions.

Current Infrastructure

The existing on-premises setup for FESS consists of several critical components:

- **Web Servers:**
  - 2 virtual machines (VMs) running IIS (Internet Information Services) for hosting the web application.
  - Load balancer to distribute traffic and ensure high availability.
- **Database Servers:**
  - 2 MySQL servers and 2 NoSQL (MongoDB) servers for data storage and management.
  - Backup web server to ensure data redundancy and reliability.
- **DNS Server:**
  - 1 DNS server to manage domain names and ensure smooth connectivity.
- **Security Components:**
  - Firewalls to protect against unauthorized access and cyber threats.
  - Router and switch for network management and connectivity.
  - Email server (Exchange server) for communication and alerts.
  - Active Directory for authentication and access control.
- **Processing and Monitoring:**
  - Systems in place for processing data and monitoring system performance.

Migration Goals and Objectives

Key Objectives:

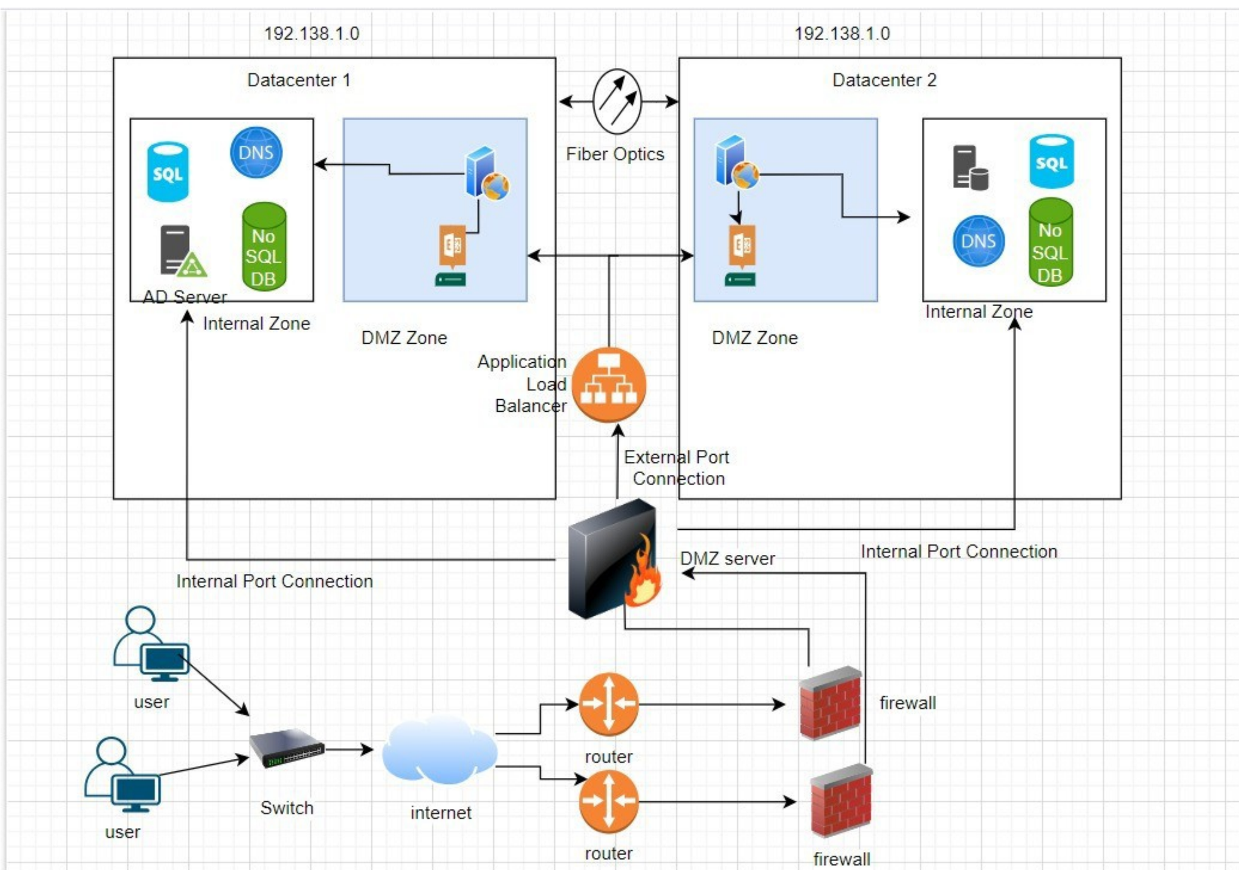
- **Cost Efficiency:** Reducing overall operational costs.
- **Performance Improvement:** Enhancing system performance and responsiveness.
- **Security Enhancement:** Strengthening security measures.
- **Strong Encryption:** Implementing robust encryption for data protection.
- **Robust Access Control:** Ensuring secure access through network segmentation.

Alignment with Organizational Strategies:

1. **Alignment with Strategic Objectives:**
   - Focus on exceptional customer experiences, operational efficiency, and innovation.
   - Integration of serverless technologies to achieve strategic alignment.
2. **Empowerment through Serverless Technologies:**
   - Rapid innovation and effective response to market dynamics.
   - Maintaining competitive edge in the retail landscape.
3. **Strategic Significance of Migration:**
   - Transition to serverless, three-tier architecture as a strategic initiative.
   - Positioning CloudFusion Solutions for long-term success in the digital era.
4. **Value Proposition:**
   - Delivering value to customers and stakeholders.
   - Driving sustainable growth and profitability through cloud migration.

Current Infrastructure Assessment

Current Architecture diagram



Comprehensive Assessment of Existing Systems

CloudFusion Solutions currently operates a robust on-premises infrastructure that supports critical business applications, including the Fusion Enterprise Security Service. The infrastructure comprises:

- **Web Servers**: 2 VMs running IIS
- **Database Servers**: 2 MySQL and 2 NoSQL (MongoDB)
- **Network Infrastructure**: Load balancer, DNS server, routers, switches
- **Security**: Firewalls, Active Directory for authentication
- **Other Components**: Email (Exchange server), backup servers, desktops, printers, storage systems, and supporting hardware (PDU, UPS)

Identified Limitations and Proposed Solutions

1. **Scalability Challenges**: The current infrastructure lacks elasticity, hindering rapid scalability in response to workload fluctuations or business growth. Moving to AWS provides elastic scalability through services like AWS Lambda for serverless computing, reducing operational complexity and costs.
2. **High Maintenance Costs**: Significant operational overhead associated with managing and maintaining hardware, including procurement, maintenance, and power consumption. AWS's cloud solutions reduce upfront hardware investments and lower operational expenses related to maintenance and power consumption.
3. **Limited Disaster Recovery**: While backup systems are in place, the infrastructure may not fully support comprehensive disaster recovery needs for uninterrupted business operations. AWS's global infrastructure enhances redundancy and disaster recovery capabilities, ensuring business continuity with automated backup and replication options.
4. **Security Concerns**: Continuous effort required to maintain robust security measures across all infrastructure layers, addressing evolving cybersecurity threats. AWS offers comprehensive security features and compliance certifications, ensuring data protection with services like Amazon Cognito for authentication and Amazon DynamoDB for secure data storage.

Cloud Service and Tool Selection

Justification for Selected Services and Tools

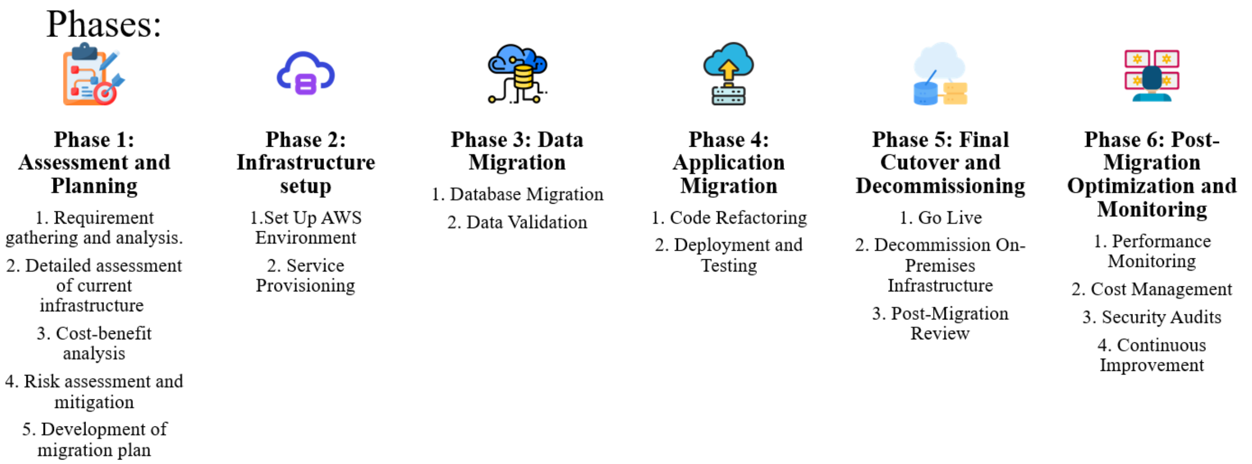
CloudFusion Solutions has carefully selected AWS (Amazon Web Services) for its cloud migration, leveraging a range of services tailored to meet the specific needs of transitioning the Fusion Enterprise Security Service to a serverless, three-tier architecture. Here's a detailed justification for each selected service and tool, emphasizing features, cost-effectiveness, and compatibility:

1. **AWS Lambda**
   - **Features**: AWS Lambda is a serverless compute service that allows executing code in response to events without provisioning or managing servers. It supports multiple

programming languages, scales automatically, and integrates seamlessly with other AWS services.
    - **Cost**: Lambda operates on a pay-per-use model, charging only for the compute time consumed during execution. This aligns with CloudFusion Solutions' goal of cost efficiency by eliminating costs associated with idle server time.
    - **Compatibility**: Lambda integrates well with various AWS services such as Amazon S3, DynamoDB, and SNS, facilitating event-driven architectures and enhancing application scalability and responsiveness.
2. **Amazon Cognito**
    - **Features**: Amazon Cognito provides authentication, authorization, and user management for web and mobile applications. It supports multiple identity providers, user pools, and federated identities, ensuring secure access control and user data management.
    - **Cost**: Cognito offers a free tier for monthly active users, with pay-as-you-go pricing beyond the free limits. This cost-effective model allows Cloud Fusion Solutions to manage authentication securely without upfront costs.
    - **Compatibility**: Integrated with AWS Amplify for seamless application deployment and AWS services for enhanced security and compliance, Amazon Cognito ensures compatibility with Cloud Fusion Solutions' existing and future applications.
3. **Amazon Simple Notification Service (SNS)**
    - **Features**: Amazon SNS is a fully managed messaging service that enables the sending of notifications and alerts to distributed systems and end-users. It supports message filtering, mobile push notifications, and integrates with other AWS services.
    - **Cost**: SNS operates on a pay-per-use pricing model, with no minimum fees or upfront commitments. This ensures cost efficiency for Cloud Fusion Solutions by paying only for the messages delivered.
    - **Compatibility**: Integrated with AWS Lambda for triggering serverless functions based on event notifications, Amazon SNS facilitates real-time communication and notification delivery, crucial for incident management and service reliability.
4. **Amazon DynamoDB (NoSQL Database Service)**
    - **Features**: Amazon DynamoDB is a fully managed NoSQL database service known for its high performance, scalability, and low latency. It is ideal for storing and retrieving security logs and operational data with predictable performance, ensuring reliability and responsiveness.
    - **Cost**: Offers flexible pricing based on provisioned throughput capacity and on-demand capacity modes, allowing Cloud Fusion Solutions to optimize costs based on application needs.
    - **Compatibility**: Integrates seamlessly with AWS Lambda for event-driven data processing and with Amazon SNS for real-time data notifications, supporting the serverless architecture's data management requirements.

Migration Strategy

Cloud Fusion Solutions will adopt a phased approach to migrate its Fusion Enterprise Security Service from on-premises infrastructure to a serverless architecture on AWS. This strategy ensures a systematic transition while minimizing disruptions to operations and maximizing the benefits of cloud scalability, cost efficiency, and enhanced security.



Security and Compliance Strategy

Cloud Fusion Solutions prioritizes robust security measures and compliance with relevant regulations in its migration to a serverless architecture on AWS. The proposed solution integrates advanced security controls and compliance frameworks to safeguard data, ensure privacy, and mitigate risks effectively.

Identity Management

Amazon Cognito:

- Utilize Amazon Cognito for authentication, authorization, and user management.
- Implement multi-factor authentication (MFA) and adaptive authentication policies to enhance identity verification.
- Integrate with existing identity providers (IdPs) and support identity federation to streamline access management across applications.

Data Protection

Encryption at Rest and in Transit:

- Implement AES-256 encryption for data at rest using Amazon DynamoDB encryption capabilities.
- Secure data transmission using TLS (Transport Layer Security) for encrypted communication between clients and AWS services.
- Ensure compliance with industry standards and regulations (e.g., GDPR, HIPAA) for data protection and privacy.

Network Security

Network Segmentation and Access Controls:

- Implement secure network architecture using Amazon VPC (Virtual Private Cloud) to isolate resources and control network traffic.
- Utilize AWS Security Groups and Network ACLs (Access Control Lists) to enforce granular access controls and restrict unauthorized access.
- Monitor and audit network traffic using AWS CloudTrail for visibility into API calls and AWS Config for compliance auditing.

Compliance with Regulations

GDPR and HIPAA Compliance:

- Implement data handling practices and encryption mechanisms to comply with GDPR requirements for data protection and privacy.
- Adhere to HIPAA regulations by implementing secure data storage and transmission practices for healthcare-related information.
- Conduct regular security assessments, audits, and penetration testing to maintain compliance and address regulatory requirements.

Incident Response and Monitoring

AWS CloudWatch and AWS Config:

- Set up continuous monitoring and logging using AWS CloudWatch for real-time visibility into application performance and security metrics.
- Configure AWS Config Rules to automate compliance checks and enforce security best practices across AWS resources.
- Establish incident response procedures and conduct tabletop exercises to prepare for and mitigate security incidents effectively.

Cost Analysis and Budget

On-Premises Infrastructure TCO

Components	Model	No of units	Hardware cost	Other cost	Cummulative cos	No of Years	Total cost
Web servers	HPE ProLiant DL380 Gen10 server	2	$13,432.00	525.60+394.20+750	$1,669.80	3	$36,882.80
Database servers	Dell EMC PowerEdge R250	4	$1,269.00	262.80+197.10+500	$959.90	3	$16,594.80
Applicationload balancer	Citrix ADC MPX5901	1	$35,291.51	315.36+236.52+3,00	$3,551.88	3	$45,947.15
DNS server	Dell PowerEdge T130	2	$1,869.00	157.68+118.26+350	$625.94	3	$7,493.64
Exchange server	Windows Server 2019 Standard	2		800+500	$1,300.00	3	$3900.00
AD servers	Lenovo ThinkServer RD450	2	$4,429.00	262.80+197.10+500	$959.90	3	$14,617.40
Router	JE468A HP Wireless Cable/DSL Firewall Router	1	$232.33	10.51+150	$160.51	3	$713.86
switch	HPE Aruba JL677A – 28-Port – Managed 10 Gigabit Ethernet Switch	1	$2,823.99	52.56+39.42+750	$841.98	3	$5,349.93
Firewall	SonicWall TZ270W	2	$787.99	650+15.77+300	$965.77	3	$7,370.60
DMZ server	ThinkSystem SR860 V2 Mission-Critical Server	1	$10,100.78	525.60+394.20+2,00	$965.77	3	$12,998.09
						SUM	$1,47,968.27

- **Hardware and Infrastructure**: $147,968.27 for 3 years
- **Personnel Costs**: $90,000 annually
- **Total for 3 Years**: $90,000 * 3 + $147,968.27 = $417,968.27

AWS Cloud TCO

Estimate summary			
Upfront cost	Monthly cost	Total 12 months cost	Currency
780	1775.18	22082.16	USD
		* Includes upfront cost	

Detailed Estimate				
Service	Upfront	Monthly	First 12 m	Currency
Amazon RDS for MySQL	0	389.75	4677	USD
DynamoDB provisioned capacity	780	124.05	2268.6	USD
Application Load Balancer	0	74.25	891	USD
Network Address Translation (NAT	0	66.6	799.2	USD
Public IPv4 Address	0	3.65	43.8	USD
Standard topics	0	72.52	870.24	USD
AWS Web Application Firewall (WA	0	540	6480	USD
AWS Amplify	0	10.73	128.76	USD
Amazon Cognito	0	25.01	300.12	USD
Amazon EC2	0	456.104	5473.25	USD
AWS Lambda	0	12.52	150.24	USD

- **All Services Cumulative Cost**: $22,082.16 * 3 + $780 = $66,026.48
- **Data Transfer**: 10 TB outbound data transfer cost: $10,800 annually (optional)
- **Operational Costs**: $50,000 annually, totaling $150,000 for 3 years
- **Total for 3 Years**: $217,026.48

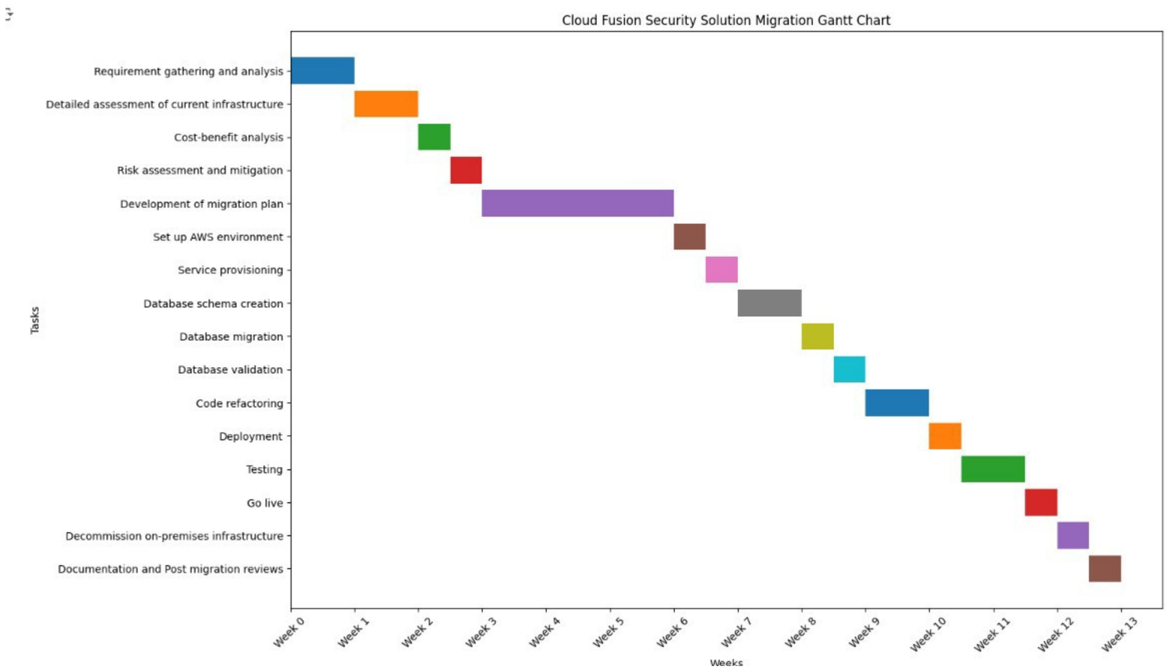
Savings and ROI

- **Savings**: Total savings from migrating to AWS: $417568.27 (On-Prem TCO) - $217026.48 (AWS TCO) = **$200541.79** over 3 years.
- **Return on Investment (ROI):** ($200541.79 / $ 417568.27) * 100 **= 48%.**

This TCO analysis demonstrates significant cost savings and a positive ROI by migrating from on-premises infrastructure to AWS serverless architecture. The savings primarily come from reduced hardware costs, lower operational expenses, and improved resource utilization in the cloud environment.

Project Management and Timeline

Cloud Fusion Solutions has outlined the following timeline for the migration project:



Team Profile

➢ Project manager

- Coordinate activities and oversee project.
- Develop and maintain plans, schedule, and budget.
- Facilitate communication and manage risks.
- Keep documentation up-to-date.

➢ Database Migration Specialist

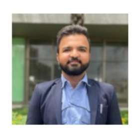
- Plan and execute database migrations.
- Ensure data integrity, security, and availability.
- Optimize database performance.
- Validate and test post-migration.
- Maintain database documentation.

➢ Cloud architect

- Design cloud architecture and infrastructure.
- Select appropriate technologies.
- Ensure scalability, security, and cost-efficiency.
- Maintain architecture documentation.
- Collaborate on deployment strategies.

➢ Application Developer

- Refactor and adapt applications for cloud.
- Test, debug, and ensure performance.
- Collaborate for seamless integration.



➢ DevOps Engineer

- Set up and maintain CI/CD pipelines.
- Automate deployment and configuration.
- Monitor system performance and manage resources.
- Implement security practices and ensure compliance.
- Support deployment and operations.

Risk Assessment and Mitigation

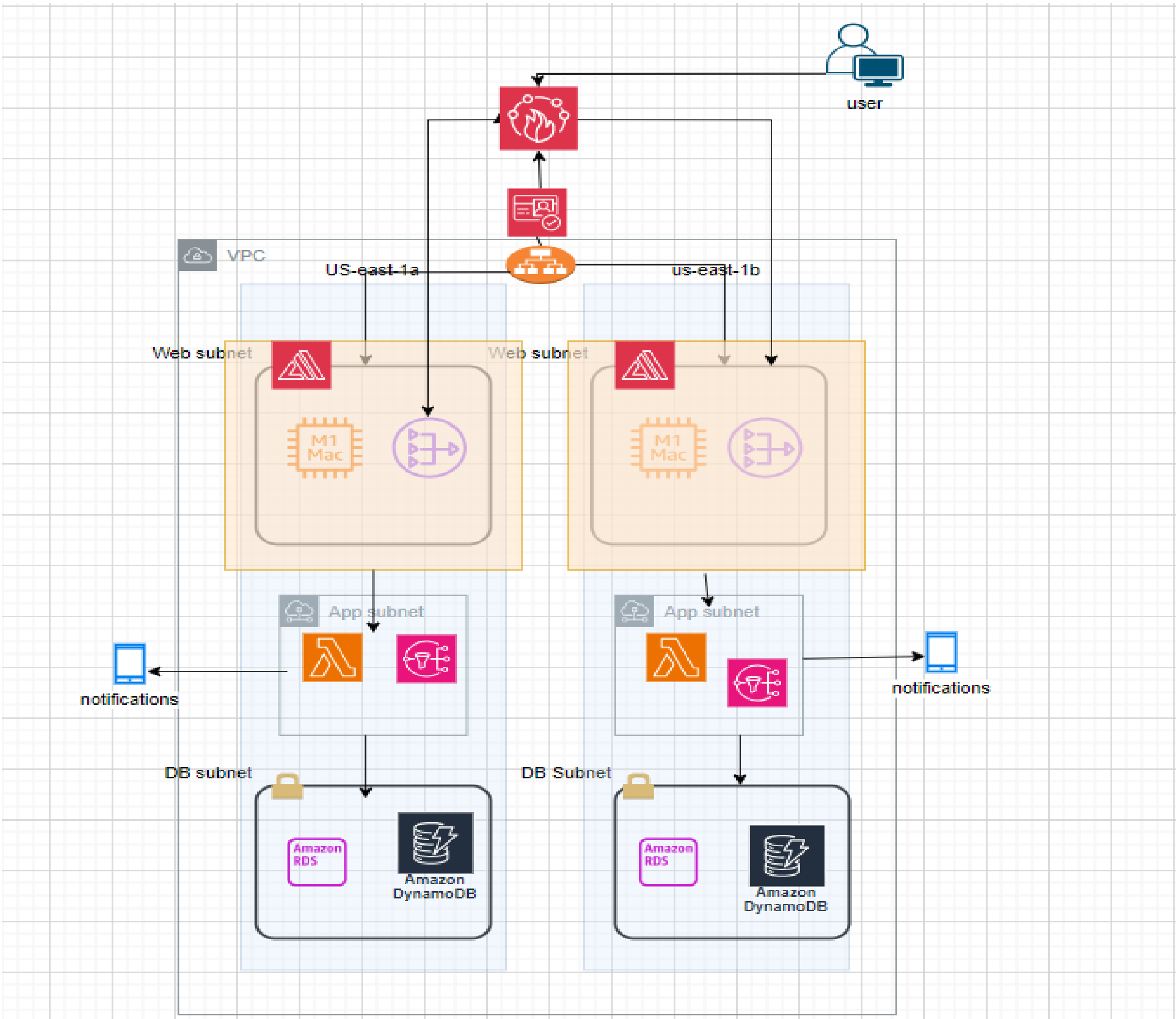
Key Risks

1. **Data Security and Privacy**: Potential breaches during data migration and integration phases.
2. **Operational Disruption**: Downtime or performance issues during application deployment and cutover.
3. **Compliance Issues**: Failure to meet regulatory requirements during and post-migration.

Mitigation Strategies

1. **Data Security**: Implement encryption at rest and in transit, conduct regular security audits, and use AWS security tools (e.g., AWS Inspector).
2. **Operational Disruption**: Phase deployment to minimize downtime, conduct thorough testing, and maintain rollback plans.
3. **Compliance Issues**: Adhere to GDPR, HIPAA, and other regulatory standards, conduct compliance audits, and implement necessary controls.

Cloud Architecture Diagram



Implementation